NVIDIA DLSS 5 and Generative Neural Rendering: Performance, Visual Fidelity, and Artistic Intent

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EE 295 - Technical Research Paper

Abstract- This paper examines the development of NVIDIA Deep Learning Super Sampling (DLSS) technology and its increasing role in real-time video-game graphics. The research begins with temporal anti-aliasing and the computational challenges of real-time ray tracing, then follows the evolution of DLSS from neural image reconstruction to Frame Generation, Ray Reconstruction, transformer-based models, and generative neural rendering. The central problem is whether generative neural rendering can improve performance and photorealism while preserving developer-created visual content and artistic intent. This study uses a literature-based comparative technical analysis and evaluates DLSS through performance, rendering workload, image quality, temporal stability, latency, generated-detail accuracy, and artistic fidelity. The paper argues that generative neural rendering has significant potential for real-time graphics; however, AI should primarily enhance developer-created content rather than substantially replace or alter the original artistic design.

Index Terms- Deep Learning Super Sampling, DLSS 5, generative neural rendering, neural rendering, real-time graphics, temporal stability, visual fidelity.

I. INTRODUCTION AND PROBLEM STATEMENT

Deep Learning Super Sampling (DLSS) uses artificial intelligence to reconstruct higher-resolution images while improving real-time graphics performance and frame rates. Real-time ray tracing increased the computational workload required for advanced lighting, reflections, and shadows, creating a need to balance image quality with playable performance. Temporal Anti-Aliasing (TAA) reuses information from current and previous frames to reduce aliasing and improve image stability, but temporal reuse can introduce blur, ghosting, and loss of fine detail [1].

Research on temporally stable neural denoising and supersampling further demonstrates the difficulty of maintaining high resolution, image quality, temporal stability, and real-time performance under limited ray budgets [4]. These challenges provide the technical motivation for neural reconstruction methods such as DLSS. Over successive generations, NVIDIA expanded AI from image reconstruction toward Frame Generation, Ray Reconstruction, transformer-based models, and generative neural rendering [5]-[7], [9], [10].

DLSS 5 introduces a broader research problem because generative neural rendering gives AI a greater role in producing the final visual appearance. This paper evaluates whether generative rendering can improve performance and photorealism without substantially altering developer-authored visual content or artistic intent. The analysis also considers the counterargument that generative AI can function as an assisting technology while humans retain creative control [13].

Existing evaluations of DLSS have primarily focused on performance, frame rates, image quality, reconstruction accuracy, and latency. The transition toward generative neural rendering creates an additional evaluation problem because performance and photorealism alone cannot determine whether generated visual information remains faithful to developer-authored content. This research therefore expands the evaluation criteria to include generated-detail accuracy, artistic fidelity, and the amount of creative control maintained by developers and artists. The contribution of this paper is to compare the technical evolution of DLSS with the changing role of AI in the graphics pipeline and evaluate whether generative neural rendering can improve real-time graphics while preserving developer-created content. The research also examines why NVIDIA continues to pursue generative neural rendering and what this transition may suggest for the future development of real-time graphics technology.

II. BACKGROUND AND LITERATURE REVIEW

A. Conventional Rendering and Temporal Anti-Aliasing

Native rendering produces an image directly at the target display resolution. A 1920 x 1080 frame contains about 2.1 million pixels, while 3840 x 2160 contains about 8.3 million pixels. Higher resolution therefore increases the amount of visual information processed per frame, especially when ray tracing is enabled. TAA addresses aliasing by accumulating temporal information across frames [1], but historical information can also produce blur, ghosting, and loss of fine detail when it no longer corresponds accurately to the current frame [1].

Fig.1 Visual comparison of texture detail with TAA disabled and enabled. Adapted from [14]

Fig.2 Comparison of no anti-aliasing, TAA, and 8x SSAA. Reproduced from [14]

Figures 1 and 2 provide visual comparisons of different anti-aliasing approaches. Figure 1 compares texture detail with TAA disabled and enabled, while Fig. 2 compares no anti-aliasing, TAA, and 8× SSAA [14]. These comparisons illustrate the trade-off introduced by temporal anti-aliasing. TAA can reduce visible aliasing and improve image stability, but temporal processing may also soften fine texture details or introduce blur. Traditional supersampling such as 8× SSAA can provide higher image quality by using additional samples, but at a significantly greater rendering cost. This trade-off between image quality and computational workload provides an important motivation for later neural supersampling techniques.

B. Neural Super Resolution and DLSS 1-2

Thomas et al. formulate neural denoising and supersampling as a multi-task learning problem in which several loss components contribute to the network's training objective [4]. The combined objective includes diffuse, specular, auxiliary, and upsampling losses, which are used together during neural-network training:

$$L := w_{diffuse} \cdot L_{diffuse} + w_{specular} \cdot L_{specular} + w_{auxiliary} \cdot L_{auxiliary} + w_{upsample} \cdot L_{upsample}$$

Each task loss is further composed of spatial and temporal loss functions:

$$L_{taski} = w_{spatiali} \cdot L_{spatiali} + w_{temporali} \cdot L_{temporali}$$

Thomas et al. determined the spatial and temporal weights by beginning with a relatively high spatial-loss contribution and reducing it until acceptable temporal stability was observed [4]. The temporal component uses an (L1)-norm based on the temporal gradient:

$$L_{temporali} = L1\,(o_i - f_w\,(o_{i-1}), r_i - f_w\,(r_{i-1}))$$

This formulation is important for real-time neural rendering because the network is not trained only to make an individual frame spatially accurate. The combined spatial and temporal objective allows the network to optimize individual-frame reconstruction while also encouraging consistency across consecutive frames [4].

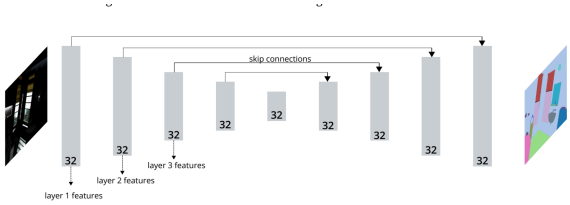

Fig.3. U-Net architecture used for temporally stable neural denoising and supersampling. Adapted from [4].

The work of Thomas et al. demonstrates an important requirement for neural reconstruction in real-time graphics: producing a high-quality individual frame is not sufficient if reconstructed details become unstable across consecutive frames [4]. This relationship between spatial image quality and temporal consistency provides a foundation for understanding neural super-resolution. Rather than relying only on conventional sampling to increase image quality, neural super-resolution uses a learned model to estimate higher-resolution visual information from lower-resolution input [2], [3].

Fig.4. Visual and performance comparison between TAA High and DLSS 2 in the tested scene. DLSS 2 increases the reported frame rate, while differences in fine wall-texture detail are visible. Adapted from [15].

As shown in Fig.4, the tested DLSS 2 configuration increases the reported average frame rate from 44 FPS with TAA High to 65 FPS, while differences in fine texture detail are visible on the wall surface [15]. This comparison illustrates an important trade-off in neural super-resolution: reducing the conventional rendering workload can improve performance, but the reconstructed output may not preserve every fine visual detail identically to the original rendering.

DLSS applies these neural reconstruction concepts to the real-time video-game graphics pipeline. The objective is not simply to enlarge a lower-resolution image, but to reconstruct a higher-resolution output using information available from the renderer. Temporal consistency is particularly important because a video game consists of a continuous sequence of frames rather than independent still images. DLSS 2 strengthened this temporal reconstruction approach by combining the current lower-resolution frame with historical information and renderer-provided motion information [5].

DLSS 2 therefore shares the concept of temporal reuse with TAA but extends it through neural reconstruction and additional information provided by the rendering pipeline [5]. At this stage of DLSS development, AI still primarily reconstructs developer-rendered information rather than independently determining the artistic appearance of the final image. This distinction between reconstruction and generation becomes increasingly important in later DLSS generations.

C. Frame Generation and Ray Reconstruction

The transition from DLSS 2 to DLSS 3 represents an important change in the role of AI. DLSS 2 primarily reconstructs a higher-resolution representation of developer-rendered information, while DLSS 3 allows the neural network to synthesize an entire intermediate frame between two conventionally rendered frames [9]. Therefore, the performance improvement is no longer achieved only by reconstructing pixels at a higher resolution; part of the

displayed frame sequence is now generated by the neural model. DLSS 3 introduced AI Frame Generation, inserting an AI-generated intermediate frame between conventionally rendered frames [9].

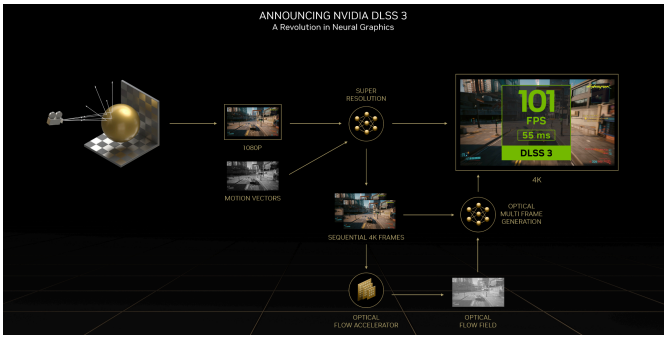

Fig.5. NVIDIA illustration of DLSS 3 Frame Generation. Adapted from [9].

Figure 5 illustrates the change introduced by Frame Generation: rather than only reconstructing a rendered frame, the neural model synthesizes additional intermediate frames [9]. This represents an important transition in the DLSS pipeline because part of the displayed frame sequence is generated rather than conventionally rendered

The process uses rendered frames, game-engine motion vectors, optical-flow information, and neural processing. This increases displayed FPS without requiring the game engine to conventionally render every displayed frame [9]. Displayed FPS, however, must be distinguished from input latency and generated-frame accuracy.

DLSS 3.5 introduced Ray Reconstruction, replacing multiple traditional denoisers with an AI-based reconstruction approach that uses spatial and temporal information to preserve ray-traced detail [10]. This represents another expansion of AI within the rendering pipeline: AI is no longer limited to reconstructing resolution or generating intermediate frames, but also reconstructs ray-traced lighting information.

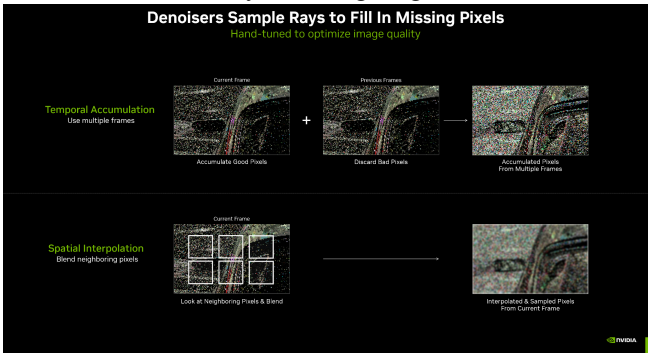

Fig.6. Illustration of traditional denoising using sampled rays to reconstruct missing pixel information. Adapted from [10].

D. Transformer Models and Generative Neural Rendering

Neural-network-based rendering research also extends beyond image upscaling and frame reconstruction. Muller et al. introduced Neural Radiance Caching (NRC), a real-time method that trains a neural network during rendering to approximate radiance for path-traced global illumination [11]. The method is designed for fully dynamic scenes without assuming fixed lighting, geometry, or materials. In their implementation, neural-cache updates and queries introduced approximately 2.6 ms of overhead at full-HD resolution while substantially reducing indirect-illumination noise with little reported bias [11]. This work demonstrates a broader direction in neural rendering: Instead of using AI only after conventional rendering to upscale an image, Neural networks can participate directly in estimating information used to construct lighting and visual appearance.

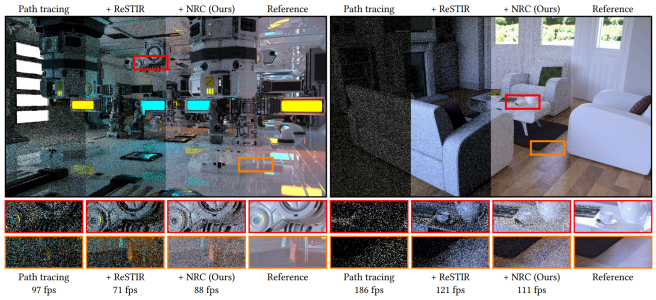

Fig.7.Comparison of path tracing, ReSTIR, Neural Radiance Caching, and reference rendering in dynamic scenes. Adapted from [11].

Although NRC is not a DLSS technology, it demonstrates the broader development of neural rendering in which learned models participate directly in computationally expensive stages of the graphics pipeline rather than functioning only as post-rendering upscalers [11]

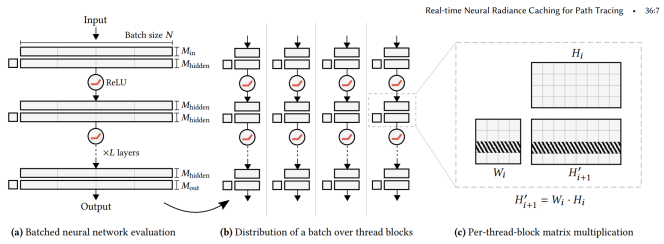

Fig.8.Neural-network evaluation and thread-block distribution used in real-time Neural Radiance Caching. Adapted from [11].

DLSS 4 continued the expansion of neural processing by introducing transformer-based models for Super Resolution and Ray Reconstruction together with Multi Frame Generation, which can generate up to three additional frames for each traditionally rendered frame [6]. The transformer-based approach increased the role of learned models in reconstructing temporally stable visual information, while Multi Frame Generation increased the amount of the displayed frame sequence produced through neural generation. NVIDIA reported improved frame-generation efficiency, although comparisons between RTX 4090 and RTX 5090 results involve different GPU generations and therefore cannot isolate algorithmic improvement alone [6].

DLSS 5 represents a different stage through generative neural rendering. NVIDIA presents the system as capable of synthesizing visual details while remaining guided by information from the game's rendering pipeline [7]. This transition creates the central research question: whether AI can take a greater role in producing visual appearance while preserving the artistic intent and creative control of human developers and artists.

Fig.9. Comparison of computer-generated input and DLSS 5 generative neural rendering output presented by NVIDIA. Adapted from [7].

III. PROPOSED APPROACH / TECHNICAL ANALYSIS

A. Comparative Technical Analysis

This research uses a literature-based comparative technical analysis rather than direct experimental testing of proprietary DLSS 5 models. Conventional rendering and TAA provide the baseline, followed by neural super-resolution and DLSS 1–2, Frame Generation in DLSS 3, Ray Reconstruction in DLSS 3.5, transformer-based reconstruction and Multi Frame Generation in DLSS 4, and generative neural rendering in DLSS 5. The comparison evaluates both performance evolution and the increasing responsibility given to neural models in producing the final visual output.

Comparing these stages is necessary because DLSS 5 cannot be evaluated only as an isolated graphics technology. Each generation represents a change in which parts of the graphics pipeline are processed by neural networks. Earlier approaches primarily reconstructed renderer-provided information, while later generations progressively introduced frame generation, ray reconstruction, and generative rendering. Examining this progression allows the research to determine whether the transition toward DLSS 5 represents a continuation of neural-assisted rendering or a more significant change in the responsibility given to AI within the graphics pipeline.

B. Evaluation Criteria

Performance is evaluated through reported frames per second (FPS) and frame-generation efficiency, while rendering workload considers how neural processing reduces or supplements conventional rendering. Image quality considers sharpness, fine-detail preservation, ray-traced lighting and reflections, and visible reconstruction or generation artifacts. These criteria evaluate whether increased neural processing produces measurable performance benefits without unacceptable reductions in visual quality.

Temporal stability is evaluated through ghosting, flickering, motion artifacts, and frame-to-frame consistency. Latency is considered separately from displayed FPS because generated intermediate frames can increase the number of frames shown to the user without requiring the game engine to process new player input for every generated frame. Therefore, displayed FPS alone is not sufficient to describe responsiveness.

Finally, generated-detail accuracy examines whether synthesized visual information remains consistent with the developer-authored source, while artistic fidelity evaluates whether the resulting output preserves the intended visual style and artistic direction. These criteria become increasingly important for DLSS 5 because photorealism does not necessarily indicate that generated information accurately represents the original developer-created content.

C. Evidence and Evaluation Strategy

Peer-reviewed research is used to establish the technical foundations of temporal anti-aliasing, neural supersampling, denoising, temporal reconstruction, and neural rendering. NVIDIA primary documentation is used to describe DLSS architecture, intended operation, and reported capabilities. Independent technical studies and benchmark sources are used where available to evaluate performance, latency, image quality, and Frame Generation behavior, while independent academic and industry sources are used to examine generative-AI controversy and artistic-control claims.

This distinction is important because NVIDIA is both the developer of DLSS and a primary source for its reported capabilities. NVIDIA documentation is therefore treated as authoritative evidence of the intended architecture and claimed performance of DLSS, but not as independent validation of those claims. Comparing primary-source information with independent research allows the analysis to distinguish the intended technical benefits of DLSS from independently observed performance, limitations, and trade-offs.

IV. RESULTS / ANALYSIS AND EVALUATION

A. Performance and Image Quality

NVIDIA reports that DLSS 4 Multi Frame Generation can generate up to three additional frames for each traditionally rendered frame and reports improvements in frame-generation efficiency compared with previous generations [6]. However, these results are associated with RTX 50-series hardware and therefore combine changes in DLSS algorithms with changes in GPU architecture. Independent evaluation by Magnius and Ali investigated the feasibility of Multi Frame Generation on an RTX 4090 [12]. Comparing these results provides a way to examine whether the performance benefits associated with Multi Frame Generation depend entirely on newer hardware or whether similar neural frame-generation concepts can provide benefits on previous-generation GPUs.

Independent testing by Magnius and Ali provides additional evidence regarding the hardware requirements of Multi Frame Generation [12]. Under identical Frame Generation 2× workloads, the RTX 4090 demonstrated slightly lower average Tensor Core utilization than the RTX 5080, while the DLAA stage accounted for the majority of Tensor Core activity. Optical Flow and Frame Generation produced relatively low and stable Tensor Core utilization across different resolutions [12]. Testing on the RTX 5080 further showed that increasing Multi Frame Generation from 3× to 4× did not substantially increase average Tensor Core utilization or burst duration. These results suggest that Tensor Core throughput alone may not explain the restriction of higher Multi Frame Generation modes to newer RTX 50-series hardware [12].

B. Temporal Stability and Latency

The distinction between spatial and temporal quality is demonstrated by Thomas et al., whose neural-rendering

objective separately considers spatial reconstruction and temporal consistency [4]. A reconstruction model may therefore produce a high-quality individual image while still performing poorly across time if reconstructed details change inconsistently between consecutive frames.

Independent testing also demonstrates that increased Frame Generation can affect frame-delivery consistency. Magnius and Ali found that native rendering on the RTX 4090 at 60, 90, and 140 FPS produced tightly clustered frametimes, with deviations limited to approximately 1 ms around the expected frametime [12]. Enabling Frame Generation 2× increased frametime variance and produced repeating three-frame spike patterns. At 60 FPS, the observed sequence reached approximately 35 ms, followed by 0 ms and another approximately 35 ms frametime before returning toward the baseline [12].

Independent testing by Tom's Hardware demonstrates the distinction between displayed FPS and responsiveness. On an RTX 5090, increasing Multi Frame Generation from disabled to 2×, 3×, and 4× increased PC latency from approximately 40 ms to 47 ms, 50 ms, and 53 ms, respectively, even as the frames-to-monitor rate increased substantially [16]. This demonstrates that higher displayed FPS can improve perceived motion smoothness without producing an equivalent improvement in input responsiveness.

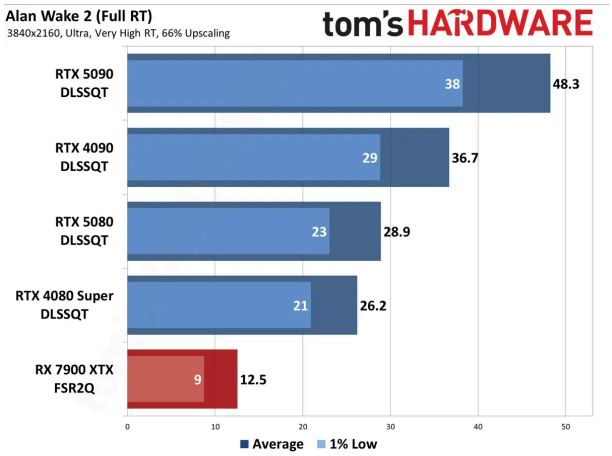

Fig.10. Benchmark from Tom's Hardware with Alan Wake 2 (Full RT). Adapted from [16]

The RTX 5080 exhibited a similar spike pattern, indicating that these spikes were not caused only by the RTX 4090's CPU-based frame pacing. However, the RTX 5080 maintained lower baseline frametime variance under comparable Frame Generation 2× conditions. The authors therefore concluded that hardware-based Flip Metering improves frame-delivery regularity compared with CPU-based pacing on the RTX 4090 [12]. These results demonstrate why displayed FPS alone is insufficient for evaluating Frame Generation: increasing the number of displayed frames does not necessarily produce equally consistent frame delivery.

C. Generated-Detail Accuracy and Artistic Fidelity

A generated visual detail can be technically plausible without being faithful to the developer-authored source. For example, a neural model may generate a realistic material, reflection, facial feature, or lighting pattern that is visually convincing but differs from the corresponding asset or artistic direction established by the developer.

Figure 9 demonstrates the type of transformation introduced by generative neural rendering. Compared with the computer-generated input, the DLSS 5 output contains additional or modified visual information intended to increase photorealism [7]. Independent analysis of the DLSS 5 reveal demonstrates why generated-detail accuracy and artistic fidelity must be evaluated separately from photorealism. Ars Technica reported criticism of the initial DLSS 5 demonstrations for producing visible changes to facial appearance, lighting, shadows, materials, and the overall visual mood of several scenes [19]. Some developers and artists argued that the generated details reduced the character of the original art direction even when the resulting images appeared more detailed or photorealistic [19]. These observations do not demonstrate that generative rendering will always reduce artistic fidelity, but they identify an important engineering requirement: generated visual information must remain sufficiently constrained by the developer-authored source if the system is intended to enhance rather than reinterpret the original design.

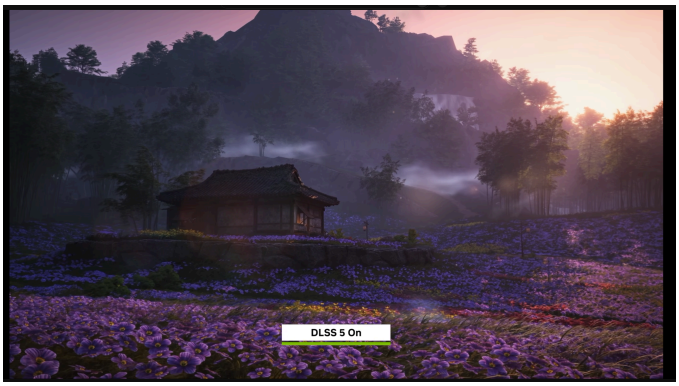

Fig.11. DLSS 5 Reveal with Where Wind Meet released in October 2026 by Official NVIDIA. Adapted from [17]

The criticism does not necessarily imply that generative neural rendering removes developer control. NVIDIA has stated that DLSS 5 provides developers with controls over the intensity, color grading, and masking of its generative effects, including the ability to disable the effect where it should not be applied [19]. Bethesda similarly stated that its art teams would continue adjusting lighting and final visual effects for its games and that the technology would remain under artist control [19]. These responses suggest that the technical question is not simply whether DLSS 5 generates visual information, but whether its developer-control mechanisms are sufficient to preserve the intended appearance of the original scene.

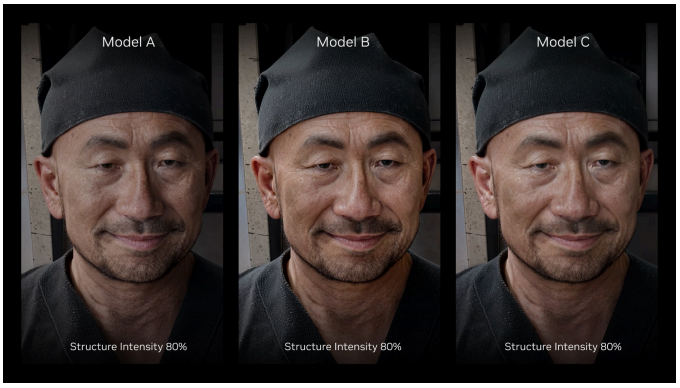

Fig. 12. Comparison of facial changes. Adapted from [18]

Figure 12 raises an important question: did the generative system preserve the identity and design characteristics that were intended to remain invariant? Answering this question should not depend on a subjective standard of beauty because different viewers may prefer different generated appearances. Instead, the evaluation should compare each generated result with the original developer-authored character. Changes in facial structure, skin texture, expression, age appearance, or other identifying features may increase perceived photorealism while simultaneously reducing fidelity to the original character design. Therefore, the important evaluation criterion is not whether one generated face appears more attractive or realistic, but whether the generative model preserves the visual characteristics intentionally established by the developer.

Capcom's position provides an important counterargument because the use of generative AI does not necessarily require transferring final artistic authority to the model [13]. If developers can constrain generated output, review its results, and determine which characteristics may be modified, generative rendering could operate as an assisting technology rather than a replacement for human artistic decisions.

The relevant evaluation criterion is therefore not simply the amount of AI-generated content. A more useful criterion is whether generated output remains predictable, temporally stable, and sufficiently constrained by developer-authored geometry, materials, lighting, and artistic direction.

Another example can be observed in *Hogwarts Legacy*, a game set within a fantasy world in which characters, environments, magical creatures, and visual effects are intentionally designed around a particular artistic direction. During the initial DLSS 5 demonstrations, one character comparison attracted criticism because the generated facial details made the character appear significantly older than the original developer-created model [19]. This example raises an important question regarding artistic fidelity: if generative neural rendering increases photorealism but changes characteristics such as perceived age, facial identity, or visual style, should the result necessarily be considered an improvement?

This question is particularly relevant because photorealism and artistic fidelity are not equivalent evaluation criteria. A fantasy game does not necessarily attempt to reproduce the physical world with complete realism; developers may intentionally use stylized characters, materials, lighting, and environments to establish a particular visual identity. Therefore, the objective of generative neural rendering should not simply be to maximize realism, but to improve visual quality while preserving characteristics that the developer intended to remain unchanged. NVIDIA's later description of the *Hogwarts Legacy* implementation states that developers can tune DLSS 5 to maintain the artistic intent and style of the game [18]. The comparison therefore demonstrates why developer control is an important technical requirement for generative neural rendering rather than only an artistic or ethical concern.

Fig.13. Hogwarts Legacy character changed. Adapted by [19]

Figure 13 illustrates why DLSS 5 has raised concerns among players and developers regarding the preservation of video-game art direction. Although the generated output may appear more photorealistic, changes to facial features and visual style may cause the final image to differ from the developer-authored design.

TABLE I
TECHNICAL EVOLUTION OF NVIDIA DLSS

Stage	Primary AI function	Main trade-off	Performance role	Image role
DLSS 2	Temporal neural reconstruction	Reconstruction artifacts	Reduce native rendering demand	Reconstruct high-resolution output
DLSS 3	Frame Generation	Latency / frame artifacts	Increase displayed FPS	Generate intermediate frames
DLSS 3.5	Ray Reconstruction	Reconstruction accuracy	Improve RT pipeline	Reconstruct ray-traced information
DLSS 4	Transformer + Multi Frame Generation	Greater neural dependency	Generate multiple displayed frames	Improved reconstruction
DLSS 5	Generative neural rendering	Source / artistic fidelity	Augment conventional rendering	Synthesize visual appearance

V. DISCUSSION AND FUTURE WORK

A. Reconstruction Versus Generation

Reconstruction primarily attempts to recover or estimate visual information from renderer-provided data, whereas generation gives the neural model greater responsibility for synthesizing visual information that contributes to the final appearance. DLSS 5 complicates this distinction because generative rendering can synthesize details while remaining guided by the game rendering pipeline [7]. The boundary between enhancement and alteration may therefore depend less on whether AI generates information and more on whether the generated information remains faithful to developer-authored content.

B. Engineering Trade-offs

The development of DLSS demonstrates that improvement in one graphics metric can introduce trade-offs in another. Neural reconstruction can reduce conventional rendering workload, and Frame Generation can increase displayed FPS [6], [9], but performance alone does not represent the complete gaming experience. Generated and reconstructed frames must also remain temporally stable, visually accurate, and responsive. Likewise, greater photorealism does not necessarily imply greater source fidelity if generated materials, lighting, faces, or environmental details differ from the developer-authored design.

Multi Frame Generation should not be evaluated as though generated FPS and conventionally rendered FPS are equivalent measurements. Generated frames can substantially improve perceived motion smoothness, while engine simulation rate and input responsiveness remain dependent on the underlying rendered frame rate.

AI assistance provides one possible way to balance these objectives. Capcom's position that generative AI can support development while maintaining human involvement in creative decisions [13] suggests that AI generation and human artistic control are not necessarily mutually exclusive. From an engineering perspective, the objective should be measurable performance and image-quality improvement while preserving predictable output and sufficient developer control.

C. Limitations and Future Research

DLSS is proprietary, so complete model architecture, training data, training methods, and internal implementation details are not public. Independent DLSS 5 evaluation is also more limited than for earlier generations, and results may vary by game, rendering pipeline, art style, motion characteristics, and developer configuration. Artistic fidelity additionally contains a subjective component that cannot be represented by a single FPS, latency, or image-quality metric.

Future research should develop quantitative measures for generated-detail fidelity and temporal consistency. Generated output could be compared against developer-authored reference content to measure changes in textures, materials, facial features, lighting, and other scene characteristics. Future neural-rendering systems could also expose developer controls that specify which visual properties generative models may modify and which must remain constrained to original assets or renderer-provided information.

VI. CONCLUSION

The evidence considered in this research suggests that generative neural rendering can provide meaningful benefits to real-time graphics, but those benefits should not be evaluated through performance or photorealism alone. As DLSS has evolved from neural reconstruction toward Frame Generation, Ray Reconstruction, transformer-based models, and generative rendering, neural systems have taken increasing responsibility for information presented in the final image. Temporal stability, generated-detail accuracy, latency, source fidelity, and developer control therefore become increasingly important evaluation criteria.

The strongest evidence for this transition is the progression from temporally stable neural reconstruction and supersampling [4], to AI-generated intermediate frames and ray-traced reconstruction [9], [10], and finally to NVIDIA's renderer-guided generative neural rendering approach in DLSS 5 [7]. Together, these developments demonstrate that the role of AI has expanded from primarily reconstructing renderer-provided information toward synthesizing a greater portion of the visual output.

Generative AI is not necessarily limited to reconstruction in order to remain compatible with human artistic control. Generation can remain useful when its output is sufficiently constrained by renderer-provided information and developer-defined artistic direction. Based on the evidence examined, generative AI should primarily function as an enhancement to developer-created graphics rather than an unconstrained replacement for human artistic decisions.

The most important future research need is the development of quantitative methods for measuring generated-detail accuracy, temporal consistency, and source fidelity across different games and visual styles. Such metrics, combined with stronger developer-control mechanisms, could help determine whether future generative rendering systems improve performance and visual quality while preserving the artistic intent of human creators.

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